

Chapter 1

Computing Numbers Wisely

Opening Challenge

Without using a calculator, compute

$$9998^2 - 9997^2.$$

One approach is to square both numbers and subtract. That works, but it requires a great deal of unnecessary computation. A better approach is to notice a pattern:

$$a^2 - b^2 = (a - b)(a + b).$$

Therefore,

$$9998^2 - 9997^2 = (9998 - 9997)(9998 + 9997) = 19995.$$

This chapter is about learning to compute wisely. Good problem solvers do not always calculate faster. Very often, they calculate less. They look for patterns, factor expressions, cancel common factors, and replace long computations by useful formulas.

As you work through the chapter, keep asking four questions:

1. Can I simplify before I compute?
2. Is there a pattern hidden in the expression?
3. Can I solve the problem in another way?
4. What happens if the numbers are replaced by variables?

The fourth question is especially important. Solving one numerical problem is useful. Discovering a formula that solves every similar problem is mathematics.

1.1 Rational and Irrational Numbers

Rational Numbers

A **rational number** is a number that can be written in the form

$$\frac{m}{n},$$

where m and n are integers and $n \neq 0$.

For example,

$$\frac{3}{4}, \quad -\frac{5}{2}, \quad 7, \quad 0$$

are rational numbers, because

$$7 = \frac{7}{1}, \quad 0 = \frac{0}{1}.$$

Thus every integer is also a rational number.

Every rational number has a decimal expansion that either terminates or repeats.

A **terminating decimal** has only finitely many digits after the decimal point. Examples include

$$0.5, \quad 2.375, \quad 18.0125.$$

A **repeating decimal** contains a digit or a block of digits that repeats forever. We place a bar over the repeating block. For example,

$$0.\overline{3} = 0.333333 \dots,$$

$$0.\overline{27} = 0.27272727 \dots,$$

and

$$4.18\overline{52} = 4.1852525252 \dots.$$

From a Repeating Decimal to a Fraction

Let us begin with a question.

Example 1.1. Express $0.\overline{7}$ as a fraction.

Solution. Let

$$x = 0.\overline{7}.$$

Multiplying by 10 moves the decimal point one place:

$$10x = 7.\overline{7}.$$

Subtracting the first equation from the second gives

$$10x - x = 7,$$

so

$$9x = 7$$

and

$$x = \frac{7}{9}.$$

Therefore,

$$0.\overline{7} = \frac{7}{9}.$$

Example 1.2. Express $0.\overline{36}$ as a fraction.

Solution. Let

$$x = 0.\overline{36}.$$

Since the repeating block has two digits, multiply by 100:

$$100x = 36.\overline{36}.$$

Subtracting gives

$$99x = 36.$$

Hence

$$x = \frac{36}{99} = \frac{4}{11}.$$

The same reasoning shows that

$$0.\overline{ab} = \frac{ab}{99}$$

when ab denotes a two-digit block, and

$$0.\overline{abc} = \frac{abc}{999}$$

when abc denotes a three-digit block.

Can You Generalize? What fraction is equal to

$$0.\overline{a_1a_2\cdots a_k}?$$

The denominator should contain k digits, all equal to 9.

Example 1.3. Express $0.21\overline{5}$ as a fraction.

Solution. Let

$$x = 0.215555\cdots.$$

First move past the nonrepeating digits:

$$100x = 21.5555\cdots.$$

Then move one more place so that the repeating parts line up:

$$1000x = 215.5555\cdots.$$

Subtracting gives

$$900x = 194,$$

so

$$x = \frac{194}{900} = \frac{97}{450}.$$

When Does a Fraction Terminate?

Suppose that m/n is written in lowest terms. Its decimal expansion terminates if and only if the denominator has no prime factors other than 2 and 5. In other words,

$$n = 2^p 5^q$$

for some nonnegative integers p and q .

Why do the primes 2 and 5 appear? A terminating decimal can be written with a denominator 10^k , and

$$10^k = 2^k 5^k.$$

Therefore the reduced denominator must divide a power of 10.

For example,

$$\frac{3}{8} = 0.375, \quad \frac{7}{25} = 0.28, \quad \frac{13}{40} = 0.325,$$

because

$$8 = 2^3, \quad 25 = 5^2, \quad 40 = 2^3 \cdot 5.$$

However,

$$\frac{1}{6}, \quad \frac{7}{12}, \quad \frac{5}{21}$$

have repeating decimal expansions because their reduced denominators contain prime factors other than 2 and 5.

Example 1.4. How many positive integers less than 100 have reciprocals that are terminating decimals?

Solution. We need all numbers of the form

$$2^p 5^q < 100.$$

They are

$$1, 2, 4, 5, 8, 10, 16, 20, 25, 32, 40, 50, 64, 80.$$

Therefore there are

$$14$$

such integers.

Irrational Numbers

An **irrational number** cannot be written as a ratio of two integers. Its decimal expansion never terminates and never repeats.

Examples include

$$\sqrt{2}, \quad \sqrt{3}, \quad \sqrt{5}, \quad \pi.$$

But how do we know that $\sqrt{2}$ is really irrational? A decimal approximation such as

$$\sqrt{2} = 1.41421356 \dots$$

does not prove that the digits will never eventually repeat. We need a mathematical argument.

Why $\sqrt{2}$ Is Irrational

Suppose, for contradiction, that $\sqrt{2}$ is rational. Then we can write

$$\sqrt{2} = \frac{a}{b},$$

where a and b are positive integers with no common factor. Squaring both sides gives

$$2 = \frac{a^2}{b^2},$$

so

$$a^2 = 2b^2.$$

Therefore a^2 is even. If a were odd, then a^2 would also be odd, so a must be even. Write

$$a = 2k.$$

Substituting into $a^2 = 2b^2$ gives

$$4k^2 = 2b^2,$$

and hence

$$b^2 = 2k^2.$$

Thus b is also even.

We have proved that both a and b are even. But then they have a common factor 2, contradicting our assumption that a/b was in lowest terms. Therefore

$$\sqrt{2} \text{ is irrational.}$$

The important idea is not merely the conclusion. We assumed that $\sqrt{2}$ could be written as a fraction and discovered that this assumption forced both the numerator and denominator to be even. The contradiction proves that no such fraction can exist.

Can You Make a Conjecture?

Now look at

$$\sqrt{3}, \quad \sqrt{5}, \quad \sqrt{6}, \quad \sqrt{7}, \quad \sqrt{8}, \quad \sqrt{10}, \quad \sqrt{11}, \quad \sqrt{12}.$$

None of the numbers under the radical sign is a perfect square. This suggests a general statement.

Conjecture. If n is a positive integer that is not a perfect square, then $\sqrt{n}$ is irrational.

This conjecture is true. The proof for a general integer is most naturally based on prime factorization. We will return to this idea when we study prime factorization and divisibility in greater depth.

For example, 49 is a perfect square, so

$$\sqrt{49} = 7$$

is rational. On the other hand, 50 is not a perfect square, so our conjecture predicts that $\sqrt{50}$ is irrational. Indeed,

$$\sqrt{50} = 5\sqrt{2}.$$

1.2 Exponents and Radicals

Positive Integer Exponents

Exponents are a convenient way to write repeated multiplication. For example,

$$2^5 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2.$$

Here 2 is the base and 5 is the exponent.

Consider

$$2^3 \cdot 2^4.$$

Writing out the factors gives

$$(2 \cdot 2 \cdot 2)(2 \cdot 2 \cdot 2 \cdot 2) = 2^7.$$

The exponents add because there are $3 + 4$ factors of 2 altogether. Therefore,

$$a^m a^n = a^{m+n}.$$

Similarly, common factors cancel in a quotient:

$$\frac{a^m}{a^n} = a^{m-n}$$

when $a \neq 0$.

For a power of a power,

$$(a^m)^n = a^{mn}.$$

For example,

$$(2^3)^4 = 2^{12},$$

because four groups of three factors contain twelve factors altogether.

For products and quotients,

$$(ab)^n = a^n b^n$$

and

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}.$$

Zero and Negative Exponents

The quotient rule suggests the meaning of a zero exponent. Since

$$\frac{a^m}{a^m} = 1$$

and also

$$\frac{a^m}{a^m} = a^{m-m} = a^0,$$

we define

$$a^0 = 1$$

for $a \neq 0$.

Likewise,

$$\frac{a^3}{a^5} = a^{-2}.$$

But canceling common factors gives

$$\frac{a^3}{a^5} = \frac{1}{a^2}.$$

Therefore,

$$a^{-n} = \frac{1}{a^n}.$$

Common Mistake. A negative exponent does not make the number negative. For example,

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8},$$

not -8 .

Roots and Fractional Exponents

For a positive number a , the square root $\sqrt{a}$ is the positive number whose square is a . Thus

$$\sqrt{49} = 7.$$

The cube root $\sqrt[3]{a}$ is the number whose cube is a . Thus

$$\sqrt[3]{125} = 5.$$

More generally, the n th root of a is written

$$\sqrt[n]{a} = a^{1/n}.$$

For positive a ,

$$a^{m/n} = (\sqrt[n]{a})^m.$$

Example 1.5. Evaluate $81^{3/4}$.

Solution. Since $81 = 3^4$,

$$81^{3/4} = (3^4)^{3/4} = 3^3 = 27.$$

Example 1.6. Evaluate $32^{2/5}$.

Solution. Since $32 = 2^5$,

$$32^{2/5} = (2^5)^{2/5} = 2^2 = 4.$$

Simplifying Radicals

When simplifying a square root, look for the largest perfect-square factor. For example,

$$\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}.$$

The factorization $72 = 8 \cdot 9$ also works, but choosing the largest perfect square often produces the simplified form immediately.

Example 1.7. Evaluate $\sqrt{27 \cdot 243}$.

Solution. Since

$$27 = 3^3, \quad 243 = 3^5,$$

we have

$$\sqrt{27 \cdot 243} = \sqrt{3^8} = 3^4 = 81.$$

Example 1.8. Evaluate

$$(4^{11} + 4^{11} + 4^{11} + 4^{11})^{1/4}.$$

Solution. There are four equal terms, so

$$4^{11} + 4^{11} + 4^{11} + 4^{11} = 4 \cdot 4^{11} = 4^{12}.$$

Therefore,

$$(4^{12})^{1/4} = 4^3 = 64.$$

Can You Generalize? Suppose that r identical terms, each equal to a^m , are added. Show that

$$\underbrace{a^m + a^m + \cdots + a^m}_{r \text{ terms}} = ra^m.$$

When r is itself a power of a , this observation can make expressions with fractional exponents much easier to evaluate.

1.3 Factoring Before Computing

Large numbers often make a problem look difficult. Before calculating, look for a common factor or a familiar algebraic identity.

The most useful identities in this section are

$$a^2 - b^2 = (a - b)(a + b)$$

and

$$a^m - a^n = a^n(a^{m-n} - 1), \quad m > n.$$

Example 1.9. Simplify

$$\frac{2^{2008} - 2^{2006}}{2^{2007}}.$$

Solution. Factor 2^{2006} from the numerator:

$$\frac{2^{2006}(2^2 - 1)}{2^{2007}} = \frac{3 \cdot 2^{2006}}{2^{2007}} = \frac{3}{2}.$$

No large power was computed.

Example 1.10. Simplify

$$\frac{2^{307} - 2^{301}}{(2^3 - 1)2^{300}}.$$

Solution.

$$\frac{2^{301}(2^6 - 1)}{(2^3 - 1)2^{300}} = 2 \cdot \frac{(2^3 - 1)(2^3 + 1)}{2^3 - 1} = 2(9) = 18.$$

Complex Fractions

A fraction whose numerator or denominator contains another fraction is called a complex fraction. Work from the innermost operation outward.

Example 1.11. Simplify

$$\frac{1}{1 + \frac{1}{2 + \frac{1}{3 + \frac{1}{4}}}}.$$

Solution. Work from the bottom upward:

$$3 + \frac{1}{4} = \frac{13}{4},$$

$$2 + \frac{1}{13/4} = 2 + \frac{4}{13} = \frac{30}{13},$$

so

$$1 + \frac{1}{30/13} = 1 + \frac{13}{30} = \frac{43}{30}.$$

Therefore,

$$\frac{1}{43/30} = \frac{30}{43}.$$

1.4 Factorials

For a positive integer n , the factorial $n!$ is defined by

$$n! = 1 \cdot 2 \cdot 3 \cdots (n-1)n.$$

For example,

$$5! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120.$$

We also define

$$0! = 1.$$

Factorials grow very quickly, but expressions involving factorials often simplify without evaluating the factorials themselves.

The identity

$$n! = n(n-1)!$$

is especially useful. More generally,

$$n! = (n-k)!(n-k+1)(n-k+2) \cdots n.$$

Example 1.12. Simplify

$$\frac{9! - 8!}{9! + 8!}.$$

Solution. Factor $8!$:

$$\frac{8!(9-1)}{8!(9+1)} = \frac{8}{10} = \frac{4}{5}.$$

Example 1.13. Simplify

$$\frac{15! - 13!}{15! + 13!}.$$

Solution. Since

$$15! = 15 \cdot 14 \cdot 13!,$$

we obtain

$$\frac{13!(15 \cdot 14 - 1)}{13!(15 \cdot 14 + 1)} = \frac{209}{211}.$$

Can You Generalize? Simplify

$$\frac{(n+1)! - n!}{(n+1)! + n!}.$$

1.5 Sums and Useful Formulas

The Sum of the First n Positive Integers

Let

$$S = 1 + 2 + 3 + \cdots + (n - 1) + n.$$

Write the same sum in reverse order:

$$S = n + (n - 1) + (n - 2) + \cdots + 2 + 1.$$

Adding the two equations term by term gives

$$2S = (n + 1) + (n + 1) + \cdots + (n + 1).$$

There are n equal terms, so

$$2S = n(n + 1).$$

Therefore,

$$1 + 2 + 3 + \cdots + n = \frac{n(n + 1)}{2}.$$

This formula is more valuable than the answer to any one numerical sum. It solves infinitely many problems at once.

Example 1.14. Find

$$11 + 12 + 13 + \cdots + 100.$$

Solution.

$$11 + 12 + \cdots + 100 = (1 + 2 + \cdots + 100) - (1 + 2 + \cdots + 10).$$

Therefore,

$$\frac{100 \cdot 101}{2} - \frac{10 \cdot 11}{2} = 5050 - 55 = 4995.$$

Geometric Sums

Consider

$$1 + r + r^2 + \cdots + r^n.$$

Let

$$S = 1 + r + r^2 + \cdots + r^n.$$

Then

$$rS = r + r^2 + \cdots + r^{n+1}.$$

Subtracting gives

$$(r - 1)S = r^{n+1} - 1.$$

Thus, for $r \neq 1$,

$$1 + r + r^2 + \cdots + r^n = \frac{r^{n+1} - 1}{r - 1}.$$

Example 1.15. Find

$$1 + 2 + 2^2 + 2^3 + \cdots + 2^{10}.$$

Solution.

$$1 + 2 + 2^2 + \cdots + 2^{10} = \frac{2^{11} - 1}{2 - 1} = 2^{11} - 1 = 2047.$$

Two additional formulas are useful in later chapters:

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6},$$

and

$$1^3 + 2^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2} \right)^2.$$

We will use these formulas when appropriate. Their proofs may be studied later by mathematical induction or other methods.

1.6 Telescoping Sums and Products

A telescoping expression is one in which many intermediate terms cancel. The basic identity is

$$\frac{1}{n(n+p)} = \frac{1}{p} \left(\frac{1}{n} - \frac{1}{n+p} \right).$$

To verify it, combine the fractions on the right:

$$\frac{1}{p} \left(\frac{n+p-n}{n(n+p)} \right) = \frac{1}{n(n+p)}.$$

In particular,

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Example 1.16. Simplify

$$\frac{1}{2000 \cdot 2001} + \frac{1}{2001 \cdot 2002} + \cdots + \frac{1}{2006 \cdot 2007}.$$

Solution. Rewrite each term:

$$\frac{1}{2000} - \frac{1}{2001} + \frac{1}{2001} - \frac{1}{2002} + \cdots + \frac{1}{2006} - \frac{1}{2007}.$$

All intermediate terms cancel, leaving

$$\frac{1}{2000} - \frac{1}{2007} = \frac{7}{2000 \cdot 2007}.$$

Example 1.17. Simplify

$$\frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \frac{1}{7 \cdot 9} + \cdots + \frac{1}{99 \cdot 101}.$$

Solution. Since

$$\frac{1}{n(n+2)} = \frac{1}{2} \left(\frac{1}{n} - \frac{1}{n+2} \right),$$

the sum becomes

$$\frac{1}{2} \left(\frac{1}{3} - \frac{1}{5} + \frac{1}{5} - \frac{1}{7} + \cdots + \frac{1}{99} - \frac{1}{101} \right).$$

Therefore,

$$\frac{1}{2} \left(\frac{1}{3} - \frac{1}{101} \right) = \frac{49}{303}.$$

Example 1.18. Simplify

$$\frac{1}{1} + \frac{1}{1+2} + \frac{1}{1+2+3} + \cdots + \frac{1}{1+2+\cdots+100}.$$

Solution. Since

$$1+2+\cdots+k = \frac{k(k+1)}{2},$$

the k th term is

$$\frac{2}{k(k+1)} = 2 \left(\frac{1}{k} - \frac{1}{k+1} \right).$$

Thus the sum is

$$2 \left(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \cdots + \frac{1}{100} - \frac{1}{101} \right) = 2 \left(1 - \frac{1}{101} \right) = \frac{200}{101}.$$

Telescoping also appears in products.

Example 1.19. Simplify

$$\left(1 - \frac{1}{2^2} \right) \left(1 - \frac{1}{3^2} \right) \left(1 - \frac{1}{4^2} \right) \cdots \left(1 - \frac{1}{50^2} \right).$$

Solution. For every $k \geq 2$,

$$1 - \frac{1}{k^2} = \frac{(k-1)(k+1)}{k^2} = \frac{k-1}{k} \cdot \frac{k+1}{k}.$$

Therefore the product is

$$\left(\frac{1}{2} \cdot \frac{3}{2} \right) \left(\frac{2}{3} \cdot \frac{4}{3} \right) \left(\frac{3}{4} \cdot \frac{5}{4} \right) \cdots \left(\frac{49}{50} \cdot \frac{51}{50} \right).$$

The first parts telescope to $1/50$, and the second parts telescope to $51/2$. Thus the value is

$$\frac{1}{50} \cdot \frac{51}{2} = \frac{51}{100}.$$

What Have We Discovered?

In this chapter, we learned more than a collection of computational rules. We learned several habits that appear throughout mathematics.

1. A rational number has a terminating or repeating decimal expansion.
2. Repeating decimals can be converted into fractions by aligning and subtracting their repeating parts.
3. Exponent laws come from counting repeated factors.
4. Large powers and factorials should usually be factored before they are computed.
5. A numerical pattern often suggests a formula involving variables.
6. Long sums and products may collapse when written in telescoping form.

The main lesson is simple:

Before calculating, look for structure.

Exercises

Students have different goals. The Core Exercises contain the essential work for this chapter. Most students should complete these problems. The Additional Practice problems may be assigned when more practice is helpful. The Challenge Problems are optional and are intended especially for students who enjoy difficult problems or wish to prepare for MATHCOUNTS and the AMC 8. Not every student is expected to complete the Challenge Problems.

Part A. Core Exercises

- Express $0.\overline{4}$ as a fraction.
- Express $0.\overline{18}$ as a fraction in lowest terms.
- Express $0.12\overline{3}$ as a fraction in lowest terms.
- Determine whether each number is rational or irrational:

$$\sqrt{64}, \quad \sqrt{20}, \quad 0.\overline{125}, \quad \pi.$$

- Which of the following fractions have terminating decimal expansions?

$$\frac{7}{16}, \quad \frac{11}{30}, \quad \frac{9}{125}, \quad \frac{13}{45}.$$

- Simplify $3^7 \cdot 3^5$.
- Evaluate $16^{3/4}$.
- Simplify $\sqrt{48}$.

- Simplify

$$\frac{3^{102} - 3^{100}}{3^{101}}.$$

- Simplify

$$\frac{12!}{10!}.$$

- Simplify

$$\frac{11! - 10!}{11! + 10!}.$$

- Find $1 + 2 + 3 + \cdots + 75$.
- Find $1 + 3 + 5 + \cdots + 99$.
- Find $1 + 3 + 3^2 + \cdots + 3^7$.

- Simplify

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{19 \cdot 20}.$$

Part B. Additional Practice

16. How many positive integers less than 500 have reciprocals that are terminating decimals?

17. Express

$$0.\overline{2} + 0.0\overline{4}$$

as a fraction in lowest terms.

18. Evaluate

$$(7^5 + 7^5 + 7^5 + 7^5 + 7^5 + 7^5 + 7^5)^{1/6}.$$

19. Simplify

$$\frac{5^{2026} - 5^{2024}}{5^{2025}}.$$

20. Simplify

$$\frac{100! + 99!}{100! - 99!}.$$

Part C. Optional Challenge Problems

21. A positive integer $n < 200$ has the property that $1/n$ is a terminating decimal. What is the greatest possible value of n ?

22. Find the value of

$$\frac{2^{2002} + 2^{2001}}{2^{2001} - 2^{2000}}.$$

23. Without expanding, evaluate

$$1001^2 - 999^2.$$

24. Evaluate

$$\frac{(50! + 49!)(48! + 47!) \cdots (2! + 1!)}{(50! - 49!)(48! - 47!) \cdots (2! - 1!)}.$$

25. Let

$$S = \frac{1}{1+2} + \frac{1}{1+2+3} + \cdots + \frac{1}{1+2+\cdots+50}.$$

Find S .

26. Simplify

$$\frac{2024! 2026!}{(2025!)^2}.$$

27. A number has the form $2^p 5^q$, where p and q are nonnegative integers. How many distinct such numbers are less than 1000?

28. Evaluate

$$\prod_{k=2}^{20} \left(1 - \frac{1}{k^2}\right).$$

Part D. Beyond the Chapter

These questions are optional. Their purpose is to help you move from particular examples to general formulas.

29. Prove that

$$0.999999 \dots = 1.$$

30. Derive a formula for

$$1 + r + r^2 + \dots + r^n.$$

31. Prove that

$$\frac{1}{n(n+p)} = \frac{1}{p} \left(\frac{1}{n} - \frac{1}{n+p} \right).$$

Then use the identity to simplify

$$\frac{1}{a(a+p)} + \frac{1}{(a+p)(a+2p)} + \dots + \frac{1}{(a+(m-1)p)(a+mp)}.$$

32. Find a general formula for

$$\frac{(n+1)! - n!}{(n+1)! + n!}.$$

Answer Key

- | | |
|----------------|---|
| 1. 4/9 | 2. 2/11 |
| 3. 37/300 | 4. rational, irrational, rational, irrational |
| 5. 7/16, 9/125 | 6. 3^{12} |
| 7. 8 | 8. $4\sqrt{3}$ |
| 9. 8/3 | 10. 132 |
| 11. 5/6 | 12. 2850 |
| 13. 2500 | 14. 3280 |
| 15. 19/20 | 16. 23 |
| 17. 4/15 | 18. 7 |
| 19. 24/5 | 20. 101/99 |
| 21. 160 | 22. 6 |
| 23. 4000 | 24. 51 |
| 25. 49/51 | 26. 2026/2025 |
| 27. 28 | 28. 21/40 |

For Problems 29–32, the expected conclusions are

$$0.999999 \dots = 1,$$

$$1 + r + r^2 + \dots + r^n = \frac{r^{n+1} - 1}{r - 1} \quad (r \neq 1),$$

$$\sum_{j=0}^{m-1} \frac{1}{(a + jp)(a + (j + 1)p)} = \frac{1}{p} \left(\frac{1}{a} - \frac{1}{a + mp} \right),$$

and

$$\frac{(n + 1)! - n!}{(n + 1)! + n!} = \frac{n}{n + 2}.$$